

Template

Studentnames and studentnumbers here

2026-06-03

Set-up your environment

```
require(tidyverse)
```

```
## Loading required package: tidyverse
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.2.0
```

```
## v forcats   1.0.1      v stringr   1.6.0
```

```
## v ggplot2    4.0.3      v tibble    3.3.1
```

```
## v lubridate  1.9.5      v tidyr     1.3.2
```

```
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
require(rmarkdown)
```

```
## Loading required package: rmarkdown
```

```
require(yaml)
```

```
## Loading required package: yaml
```

```
library(tidyr)
```

```
library(writexl)
```

```
library(readxl)
```

Title Page

A new sentence Include your names

Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Include the following:

- Why is this relevant? (Smith, 2020)
- ...

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
dataset <- midwest
```

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

```
head(dataset)
```

```
## # A tibble: 6 x 28
##   PID county  state  area poptotal popdensity popwhite popblack popamerindian
##   <int> <chr>   <chr> <dbl>    <int>      <dbl>    <int>    <int>         <int>
## 1  561 ADAMS   IL    0.052   66090     1271.    63917    1702          98
## 2  562 ALEXAND~ IL    0.014   10626      759     7054    3496          19
## 3  563 BOND    IL    0.022   14991      681.   14477    429          35
## 4  564 BOONE   IL    0.017   30806    1812.   29344    127          46
## 5  565 BROWN   IL    0.018    5836     324.    5264    547          14
## 6  566 BUREAU  IL    0.05    35688     714.   35157     50          65
## # i 19 more variables: popasian <int>, popother <int>, percwhite <dbl>,
## #   percblack <dbl>, percamerindian <dbl>, percasian <dbl>, percother <dbl>,
## #   popadults <int>, perchsd <dbl>, percollege <dbl>, percprof <dbl>,
## #   poppovertyknown <int>, percpovertyknown <dbl>, percbelowpoverty <dbl>,
## #   percchildbelowpovert <dbl>, percadultpoverty <dbl>,
## #   percelderlypoverty <dbl>, inmetro <int>, category <chr>
```

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```
mean(dataset$percollege)
```

```
## [1] 18.27274
```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

Variable 2

3.3 Visualize temporal variation

3.4 Visualize spatial variation

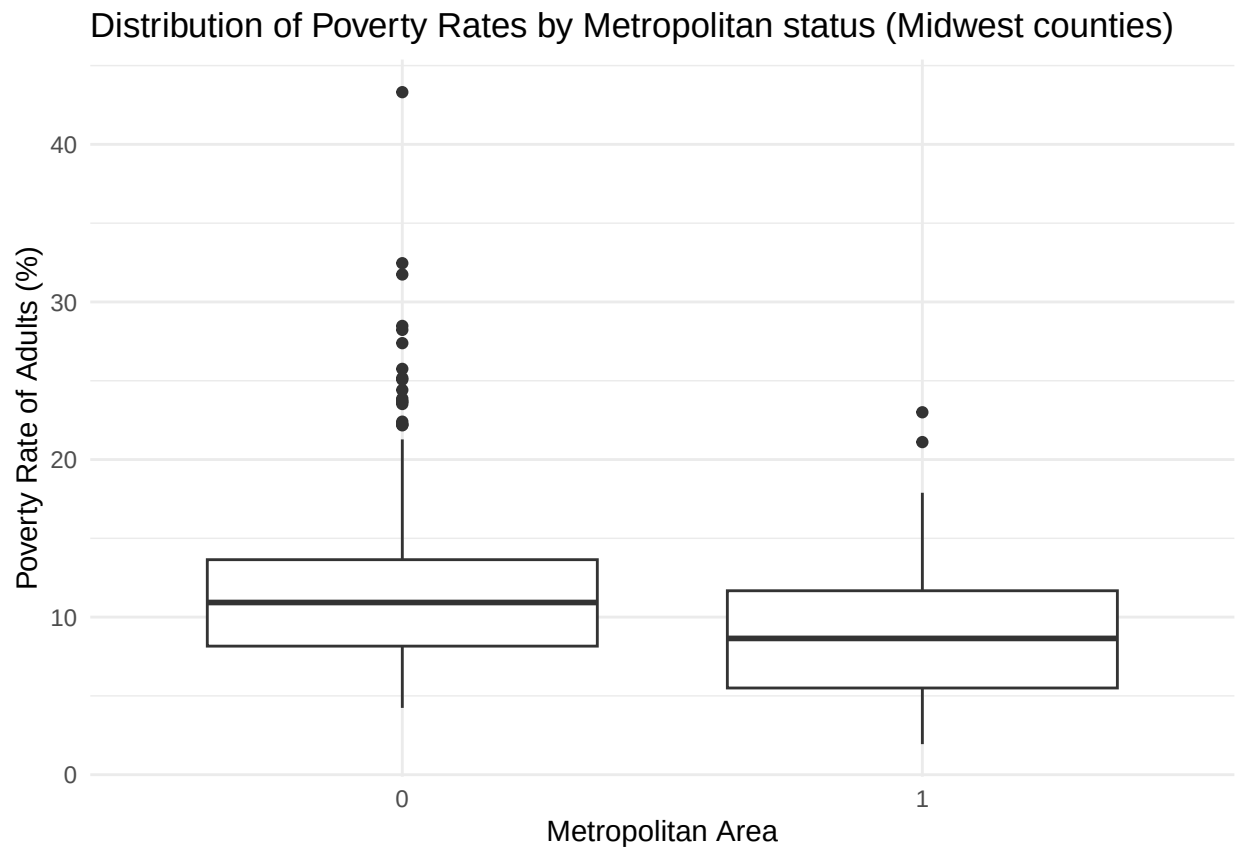
Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

```
dataset$inmetro <- dataset$inmetro %>% as.factor()
# Boxplot of poverty rate by state using the 'midwest' dataset
ggplot(dataset, aes(x = inmetro, y = percadultpoverty)) +
  geom_boxplot() +
  labs(
    title = "Distribution of Poverty Rates by Metropolitan status (Midwest counties)",
    x = "Metropolitan Area",
    y = "Poverty Rate of Adults (%)"
  ) +
  theme_minimal() +
```

```
theme(  
  legend.position = "right"  
)
```



Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.

Smith, J. (2020). My article. *Journal of Things*, 1, 1–10.